

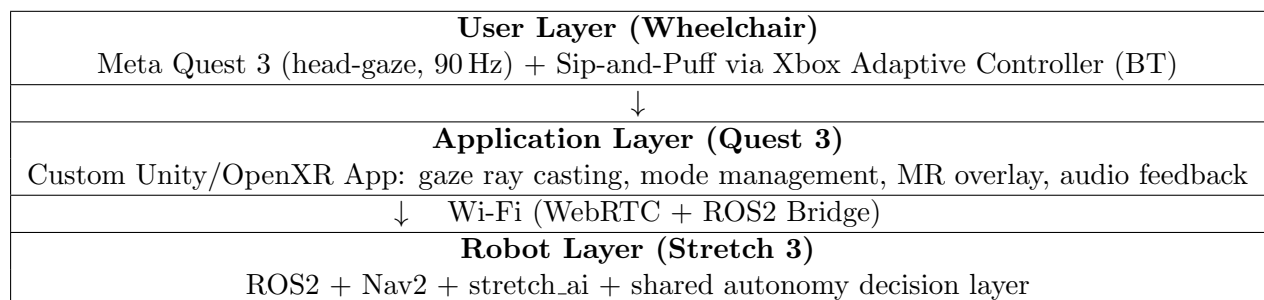
GazeBot: Head-Gaze and Sip-and-Puff Shared Autonomy for Assistive Robot Manipulation

Hello Robot Stretch 3 + Meta Quest 3 • 8 Weeks, 10 hrs/week

Het | Prof. Harshal P. Mahajan, Ph.D. | McKechnie Family LIFE Home, UIUC

System Architecture

Overview



Input Mapping

Input	Action
Head rotation (yaw/pitch)	Gaze ray direction—selects targets in robot camera view
Soft sip	Confirm / Select
Soft puff	Cancel / Back
Hard sip	Switch control mode
Hard puff	Emergency stop—immediately halts all robot motion

Control Modes

Mode	Head-Gaze Maps To	Sip Action
Look	Robot head pan/tilt (camera)	Set viewpoint
Navigate	Target destination in environment	Go to target (Nav2)
Manipulate	Object selection via gaze ray	Grasp/place (autonomous)
Menu	UI option navigation	Select preset task

Shared Autonomy Pipeline

1. **User looks at object** → head-gaze ray highlights detected object in MR overlay
2. **User sips to confirm** → `stretch_ai` identifies object via YOLO/SAM
3. **Robot plans grasp** → trajectory preview shown in MR view
4. **User sips to approve** → robot executes autonomous grasp
5. **User looks at destination** → head-gaze indicates placement location
6. **User sips to confirm** → robot navigates (Nav2) and places object

Technology Stack

Component	Technology
XR Application	Unity 2022 LTS + Meta XR SDK + OpenXR
Head Tracking	Meta Quest 3 inside-out tracking (6DoF, 90 Hz)
Sip-and-Puff	Generic switch + Xbox Adaptive Controller (Bluetooth)
Robot Control	ROS2 Humble + <code>stretch.body</code> Python SDK
Navigation	Nav2 (ROS2) with RPLIDAR A1
Object Detection	<code>stretch.ai</code> (YOLO + SAM)
Grasp Planning	<code>stretch.ai</code> / FUNMAP whole-body grasping
Video Streaming	WebRTC (via <code>stretch_web_teleop</code> infrastructure)
Audio Feedback	Bluetooth earbuds with confirmation tones

Methodology

The project follows an iterative, incremental development approach. Each week produces a functional system increment that can be demonstrated and tested in the LIFE Home environment.

Development workflow: Each component builds on the previous week’s deliverable, starting with basic robot control and progressively adding head-gaze input, sip-and-puff integration, autonomous navigation, shared autonomy manipulation, and task presets.

Evaluation: Structured task trials conducted in the LIFE Home’s home simulation area with observation cameras recording for analysis. Five standard ADL tasks evaluated across 3 trials each: (1) fetch a specified object, (2) navigate to a different room, (3) open a cabinet and retrieve an item, (4) pick up an object from the floor, (5) execute a preset “get water” sequence. NASA-TLX and System Usability Scale (SUS) administered post-evaluation.

Timeline (8 Weeks, 10 hrs/week)

Week	Phase	Key Tasks and Milestone
1	Setup & Baseline	Configure Stretch 3 with ROS2/ <code>stretch_ai</code> in LIFE Home. Set up Unity + Meta XR SDK. Get OPEN TEACH baseline running. <i>Milestone: Stretch controllable via Quest 3 hand tracking.</i>
2	Head-Gaze Control	Build Unity app reading head pose via OpenXR. Implement gaze ray casting. Stream Stretch camera into Quest 3 via WebRTC. Map head rotation to robot camera pan/tilt. <i>Milestone: User sees through robot's eyes, controls camera with head.</i>
3	SNP + Modes	Connect sip-and-puff to Xbox Adaptive Controller over Bluetooth. Map sip/puff signals to actions. Implement 4 control modes. Build MR HUD. <i>Milestone: Full hands-free input chain working across all modes.</i>
4	Navigation	Integrate Nav2—head-gaze selects target, sip confirms. Convert gaze ray + depth to 3D goal. Show path preview in MR. Test throughout LIFE Home. <i>Milestone: User looks at a point, sips, robot navigates there.</i>
5	Manipulation	Integrate <code>stretch_ai</code> object detection. Gaze ray highlights objects. Sip triggers autonomous grasp. Grasp preview in MR. “Bring to me” behavior. <i>Milestone: User looks at object, sips, robot picks up and brings it.</i>
6	Task Presets	Build preset tasks: “Fetch object”, “Open door”, “Clear path”, “Get water”. Head-gaze menu. Audio feedback. Emergency stop refinement. <i>Milestone: Complete system with preset ADL tasks and safety stops.</i>
7	Evaluation	Structured evaluation: 5 tasks $\times$ 3 trials. Baseline comparison with <code>stretch_web_teleop</code> . Administer NASA-TLX and SUS. Data analysis. <i>Milestone: Evaluation data demonstrating system viability.</i>
8	Dissemination	Record demo video in LIFE Home. Write technical report. Clean codebase, push to GitHub. Prepare lab presentation. Draft conference paper outline. <i>Milestone: Demo video, report, GitHub repo, paper outline.</i>

Required Resources

Available at LIFE Home

Resource	Status
Hello Robot Stretch (mobile manipulator)	Available (bookable)
Meta Quest / Oculus VR headset	Available in Extended Reality Lab
Home simulation area	Available for testing
Observation cameras (6× Axis M5525-E)	Available for evaluation recording
Wi-Fi infrastructure	Available

To Be Procured (Estimated: \$200–\$400)

Item	Est. Cost
Xbox Adaptive Controller	\$99
Sip-and-puff switch (3.5mm jack)	\$50–150
Gooseneck mount for SNP tube	\$20–40
Bluetooth earbuds (audio feedback)	\$30–50
Total	\$200–340

Software (All Free / Open-Source)

Unity 2022 LTS, Meta XR SDK, ROS2 Humble, `stretch.ai`, `stretch_body`, `stretch_ros2`, OPEN TEACH.